Khoa Pan



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Khoa/Mawa Making Pan

What it is used for

To heat milk or a milk-based mixture in a khoa pan until the required moisture is removed and khoa or mawa is formed. To maintain uniform heating and stirring while preventing scorching, overflow, and contamination.

Main parts

- Heating pan or vessel
- Gas burner or electric heating system
- Temperature or heat control
- Agitator or manual stirring tool
- Support frame and heat insulation
- Discharge or product collection arrangement

Before you start

1. Ensure the pan and all milk-contact surfaces are clean and suitable for food processing.
2. Inspect the pan, support frame, heating system, controls, and fuel or electrical connections.
3. Confirm that the pan is stable and the discharge area is prepared.
4. Use fresh milk or an approved milk mixture and check its quality before processing.
5. Prepare clean, dry food-grade tools and containers for collecting the khoa.
6. Wear suitable heat-resistant gloves, protective clothing, and closed footwear.

How to use

1. Place the pan securely and confirm that the heating source is switched off before loading.
2. Pour the required quantity of milk into the pan without exceeding its safe working capacity.
3. Start the heating system according to the manufacturer's instructions.
4. Heat the milk gradually and stir continuously or at the frequency required by the process.
5. Monitor the heating rate and scrape the bottom and sides using an approved food-safe tool.
6. Adjust heat to maintain controlled evaporation and prevent scorching or excessive foaming.
7. Continue heating and stirring as the milk thickens and moisture reduces.
8. Check the product texture and finish against the approved process requirements.
9. Reduce or switch off heat when the required khoa consistency is reached.
10. Transfer the hot product carefully into a clean, dry, food-grade container using suitable tools.
11. Switch off the heating source and isolate fuel or power as applicable.
12. Clean the pan and product-contact tools promptly after they have cooled sufficiently.

⚠ Safety

- Only trained personnel should operate the heating equipment.
- Avoid contact with hot milk, steam, the pan surface, and hot metal tools.
- Never overfill the pan; boiling milk can foam and overflow suddenly.
- For gas heating, check for leaks and maintain suitable ventilation.
- Keep water away from hot fat or concentrated milk residue to reduce splashing risks.
- Do not leave the heating process unattended and keep combustible materials away from the burner.

Care & storage

- Clean the pan and food-contact tools after every batch using approved food-safe procedures.
- Inspect the pan surface, support frame, burner, and controls regularly.
- Remove burnt residue using a method that does not damage the food-contact surface.
- Keep gas lines or electrical connections in safe condition.
- Arrange periodic maintenance and check temperature controls where fitted.

Common mistakes to avoid

- Heating too quickly and scorching milk at the bottom.
- Failing to stir and allowing the product to stick or burn.
- Overfilling the pan and causing overflow.
- Using unclean containers or tools that contaminate the khoa.
- Touching hot surfaces or transferring the product without suitable protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.